# *INTRODUCTION TO COMPUTER GRAPHICS*

# What is Computer Graphics?

- Course Objectives:
  - A *fundamental understanding* of the *technology underlying* the field of **Computer Images** and how this is *applied to advanced* areas such as geometric modelling, rendering and animation.
- Definition
  - Producing pictures or images using a computer
  - *Computer graphics is concerned with all aspects of producing pictures or images using a computer.*

| Synthesis: | | Analysis: |
|---|---|---|
| Computer Graphics | ⟷ | Image analysis |

- Example
  - Starship Troopers: Tango-Urilla, Death From Above

*Courtesy of Tippet Studio*

# What is Computer Graphics?

- Definition
  - Producing pictures or images using a computer
- Example
  - Starship Troopers: Tango-Urilla, Death From Above
    - On set: Pyrotechnics

*Courtesy of Tippet Studio*

# What is Computer Graphics?

- Definition
  - Producing pictures or images using a computer
- Example
  - Starship Troopers: Tango-Urilla, Death From Above
    - Bugs:

*Courtesy of Tippet Studio*

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  - Starship Troopers: Tango-Urilla, Death From Above
    - The Final Stage: Burning the bugs

*Courtesy of Tippet Studio*

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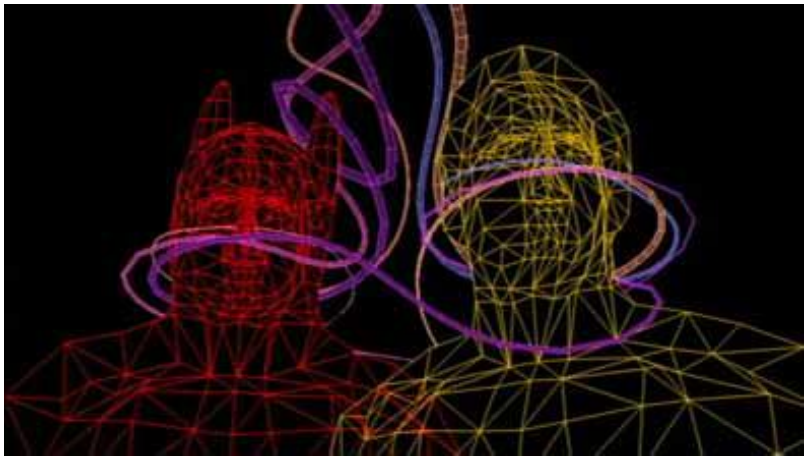

*Courtesy of Buf Compagnie*

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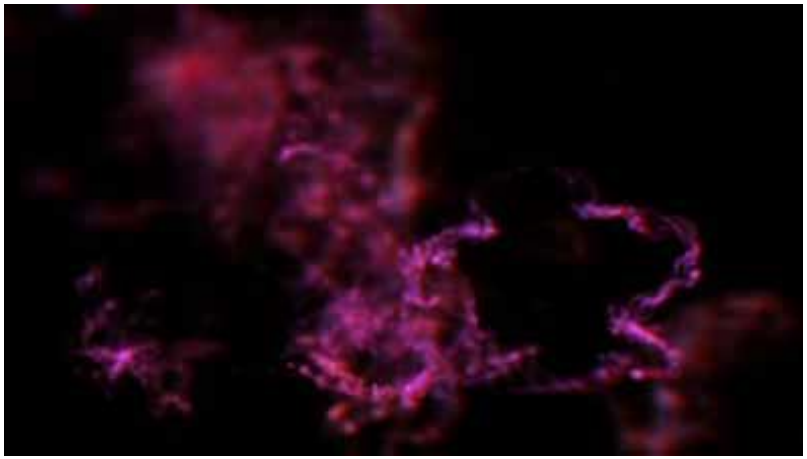

*Courtesy of Buf Compagnie*

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*Courtesy of Buf Compagnie*

# What is Computer Graphics?

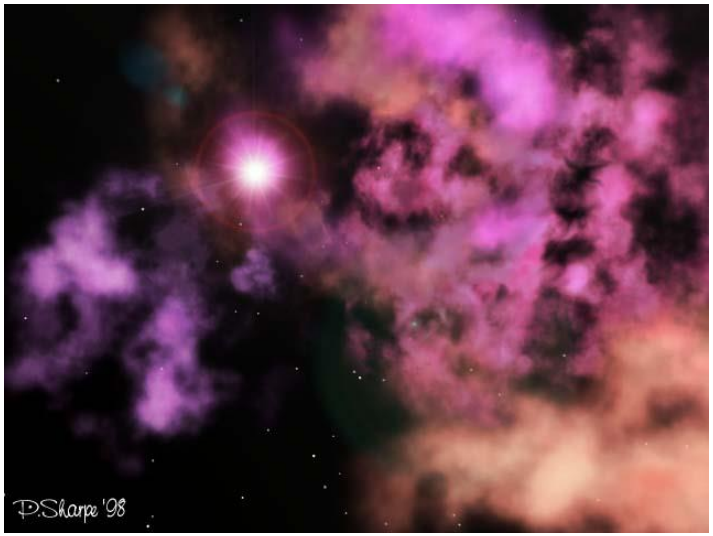
- Imaging
  - Representing 2D images
- Modeling
  - Representing 3D objects
- Rendering
  - Constructing 2D images from 3D models
- Animation
  - Simulating changes over time

# Applications

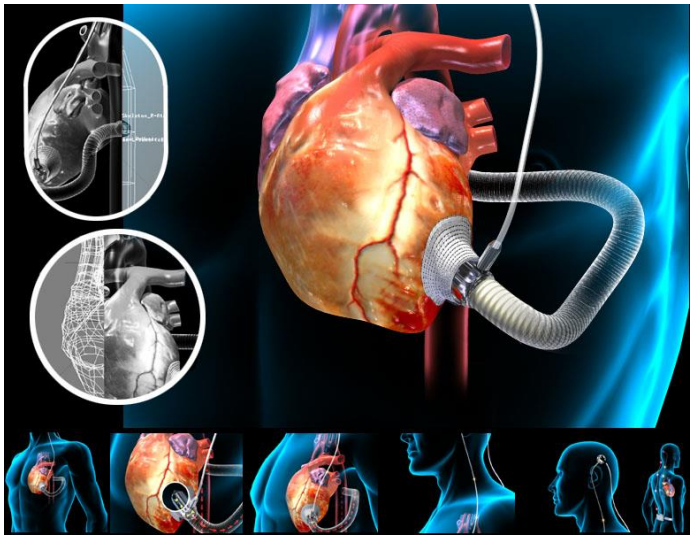
- Display of Information
- Design
  - CAD in architecture, chip design, etc.
- Simulation & Animation
  - Weather report on TV.
  - Entertainment: Movies, games etc.
- Visualization
  - Scientific data, medical data, etc.
- Computer Art
- Entertainment
- Presentation
- Education &Training
- Image Processing
- Graphics User Interface
  - Windows systems, etc.

# Display of Information

- Graphics for Scientific, Engineering, and Medical Data



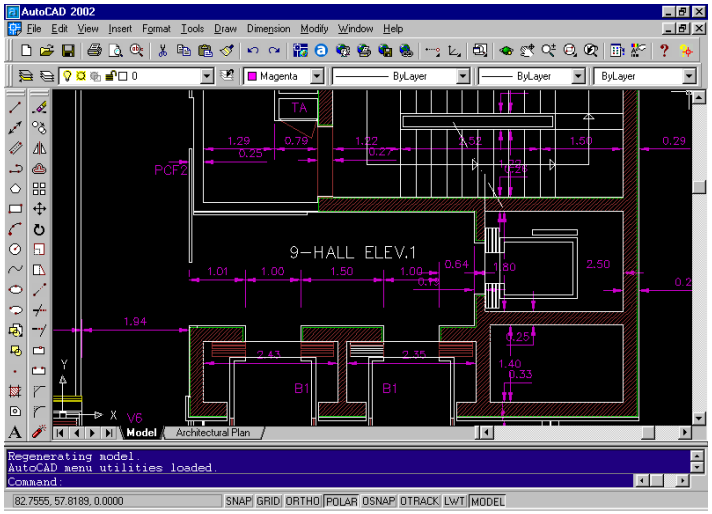

Nebula



Medical Image

# Design

- Graphics for Engineering and Architectural System
- Design of Building, Automobile, Aircraft, Machine etc.




AutoCAD 2002

Interior Design

# Simulation

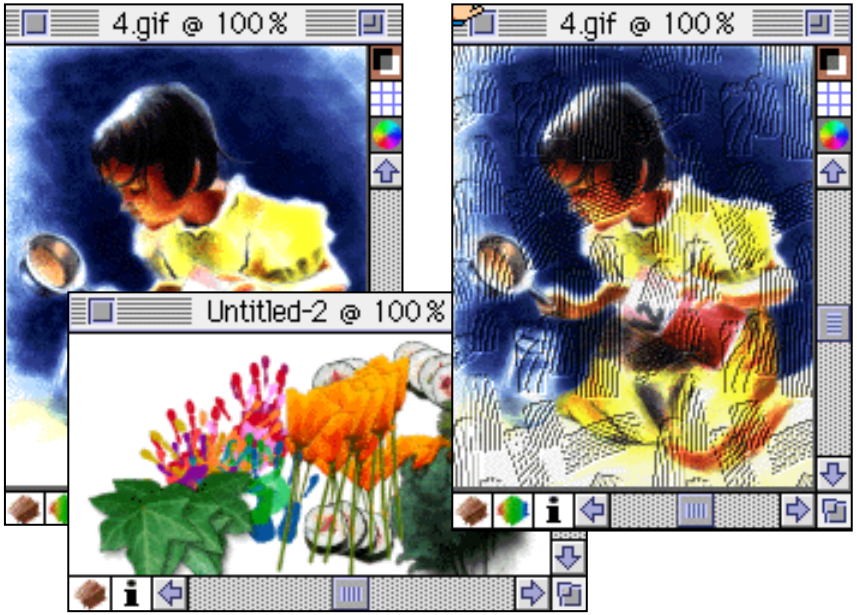
- Computer-Generated Models of Physical, Financial and Economic Systems for Educational Aids

Flight Simulator

Mars Rover Simulator

# Computer Art

- Graphics for Artist




Metacreation Painter

# Entertainment

- Graphics for Movie, Game, VR etc.

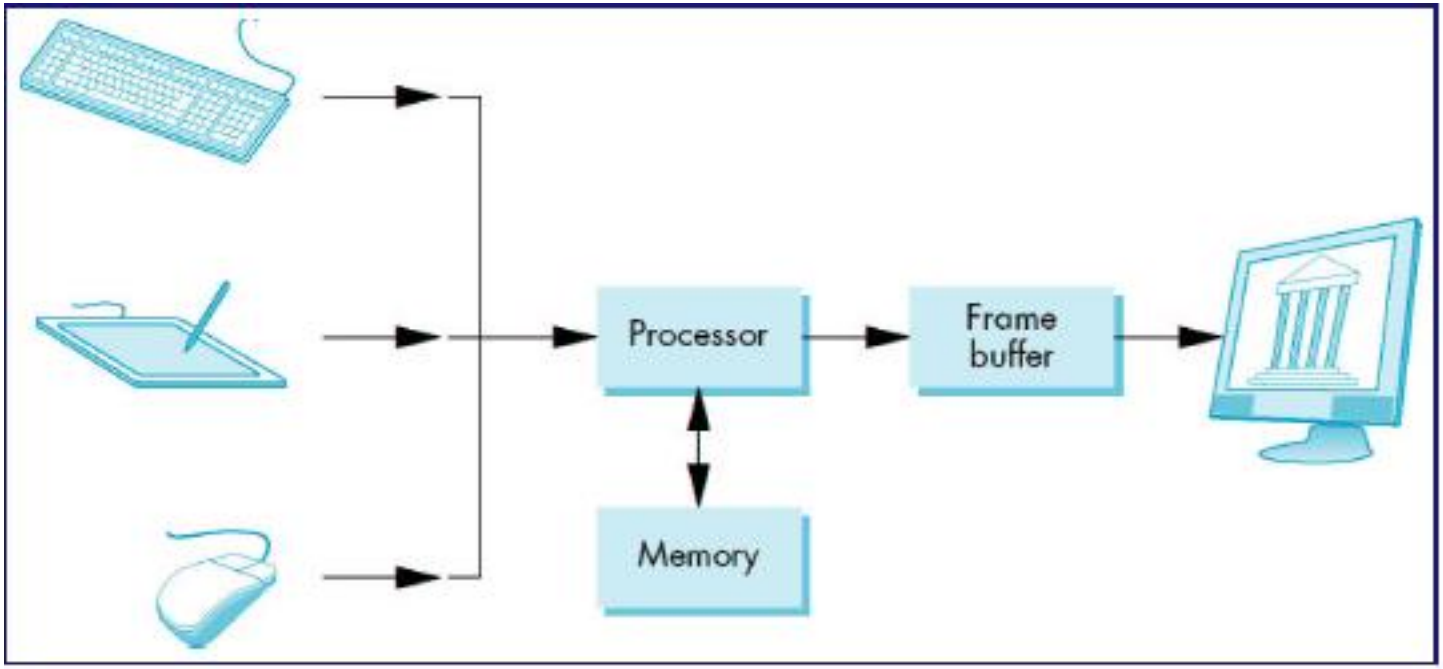
Final Fantasy

Online Game

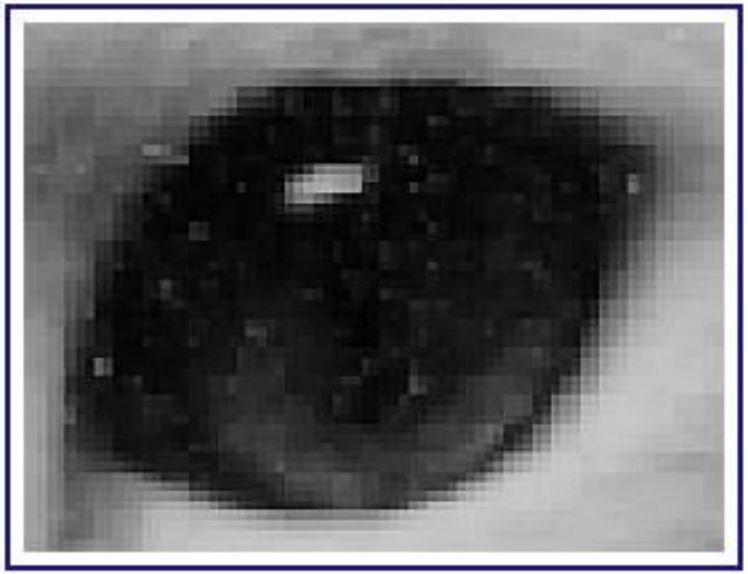
# What's Our Scope?

- Not a Tutorial on Commercial Software
  - 3DMax, Maya, Photoshop, etc.
- Not about Graphics Business
  - 3D online-game, E-commerce, etc.
- Graphics = Algorithm for Visual Simulation
  - Imaging, Modeling, Rendering, Animation

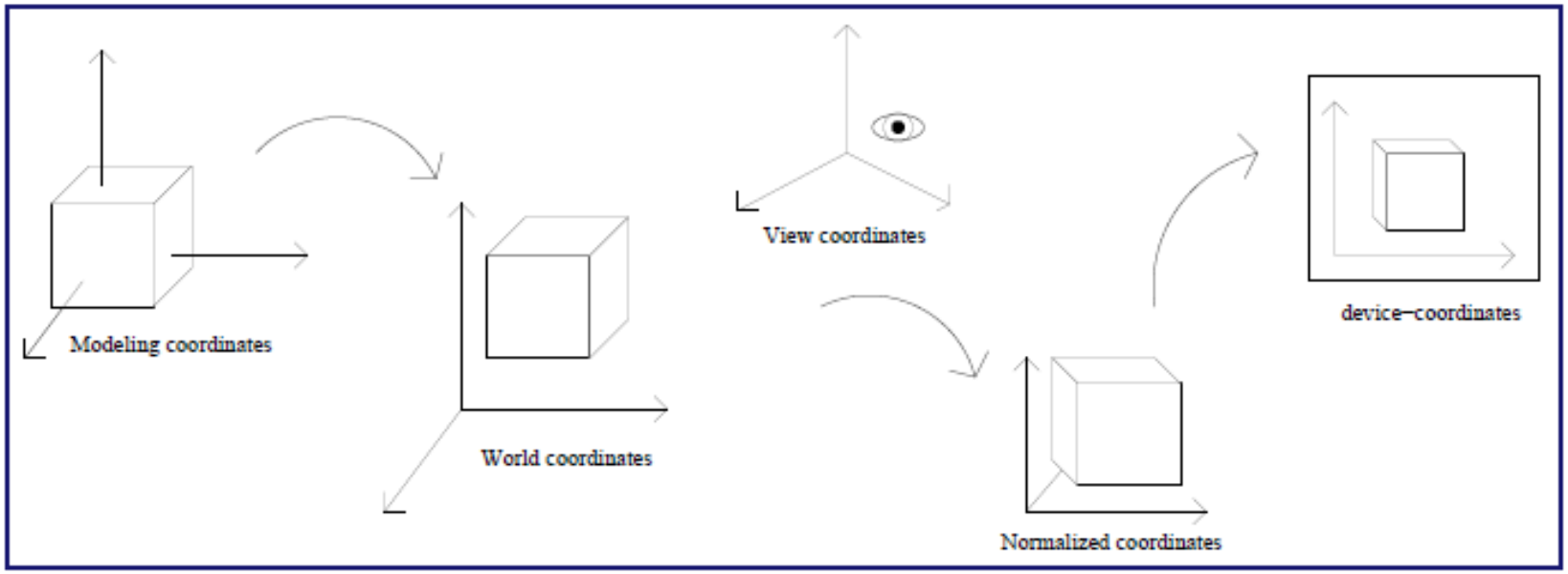
# A Graphics System



# Images: Physical & Synthetic



# Coordinates




$$(x_m c, y_m c, z_m c) \rightarrow (x_w c, y_w c, z_w c) \rightarrow (x_v c, y_v c, z_v c)$$

$$\rightarrow (x_n c, y_n c, z_n c) \rightarrow (x_d c, y_d c)$$

# Computer Graphics API

In general graphics API's include functions for:

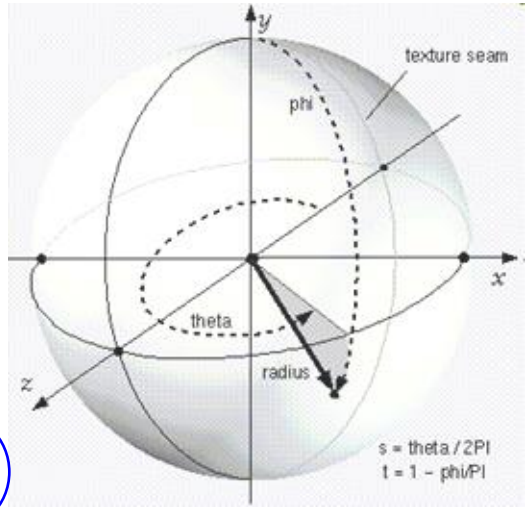
- Drawing primitives
- Setting attributes
- Viewing
- Transformation
- Input

# *Imaging*

- Warping
- Metamorphosis
- Non-Photorealistic Rendering

# Modelling

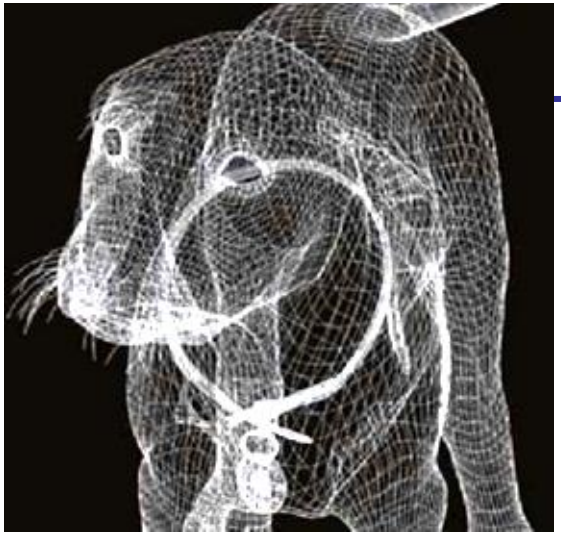
CONCEPT

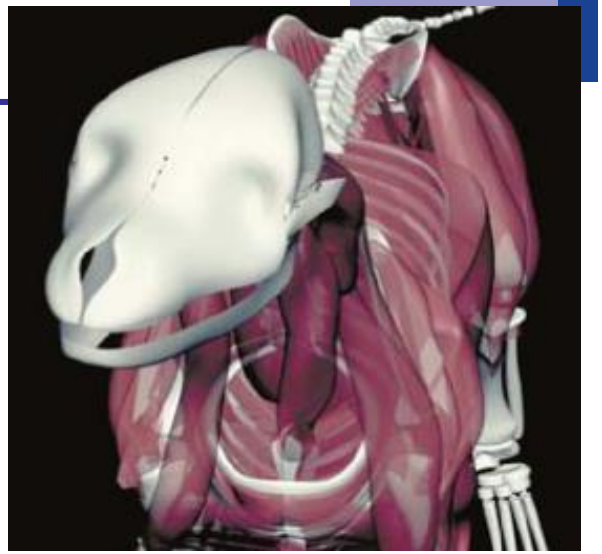



MODEL

IMAGE

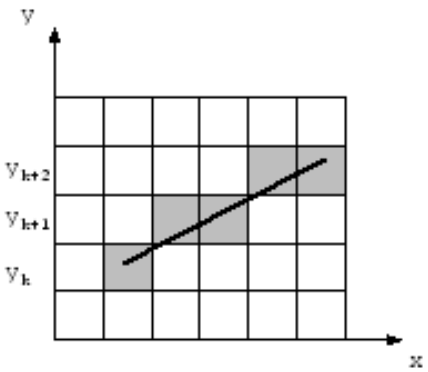
MODELS (in computer graphics): A Digital, Descriptive or Mathematical representation of a scene or object that can be stored (efficiently) in computer memory.

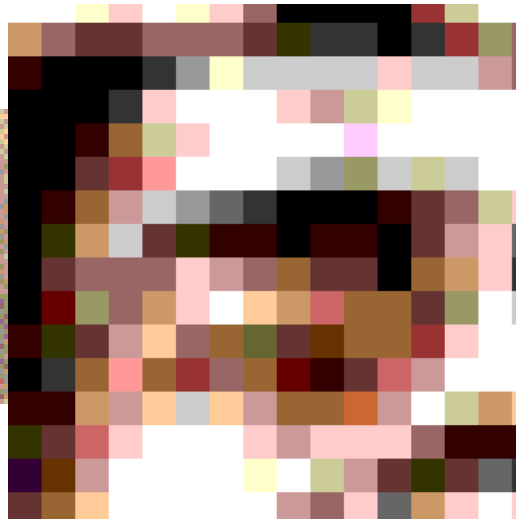




# Basic Issues

- 2D Issues
  - Line Drawing
  - Clipping
  - Polygon Filling
  - Curves
- Rasterization!
- Discretization.



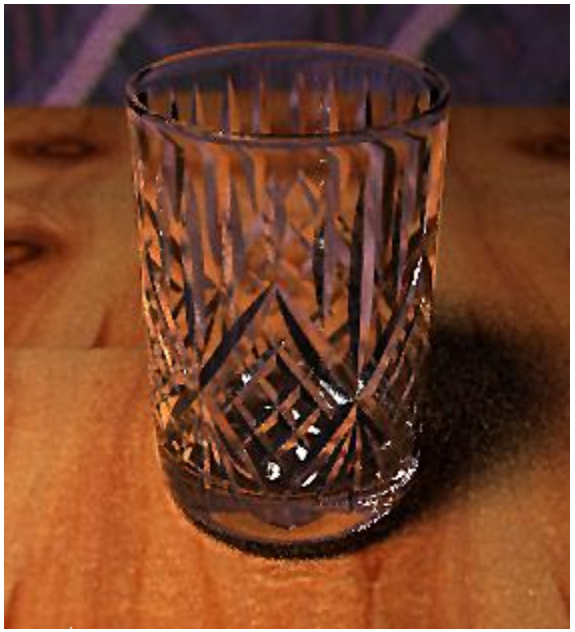


# Digital Images

- An image is a visual output of data stored in terms of a numeric, recordable elements.
- Most digital images we use today are output in terms of a regular grid of pixels, referred to as a raster.
- Intermediately the image can be stored in the form of lines, curves and filled areas, referred to as vector primitives.
- All of these are eventually represented in a numerical way.

# Application Areas

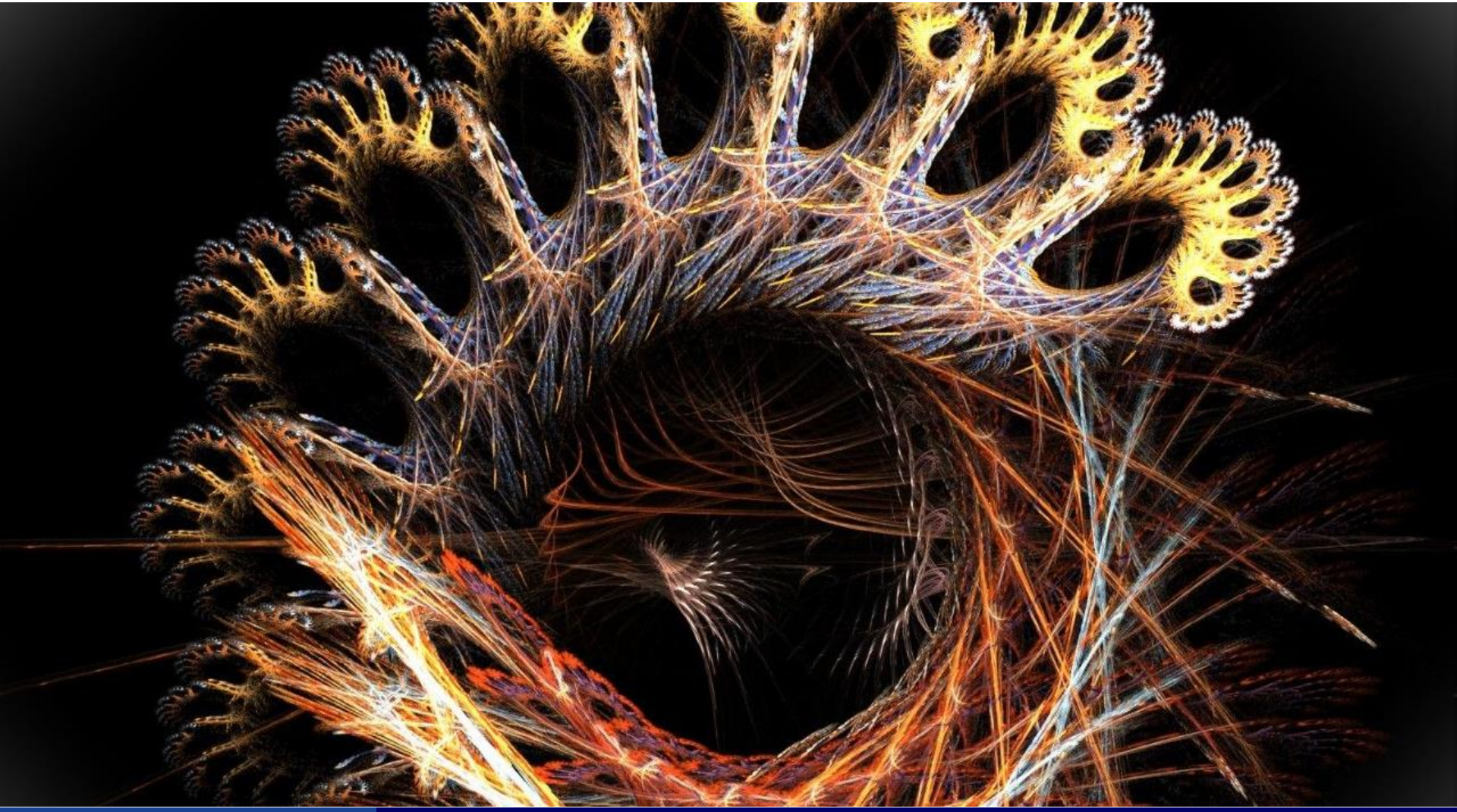
# Image Synthesis



# Computer Animation

# Aiding Traditional Animation

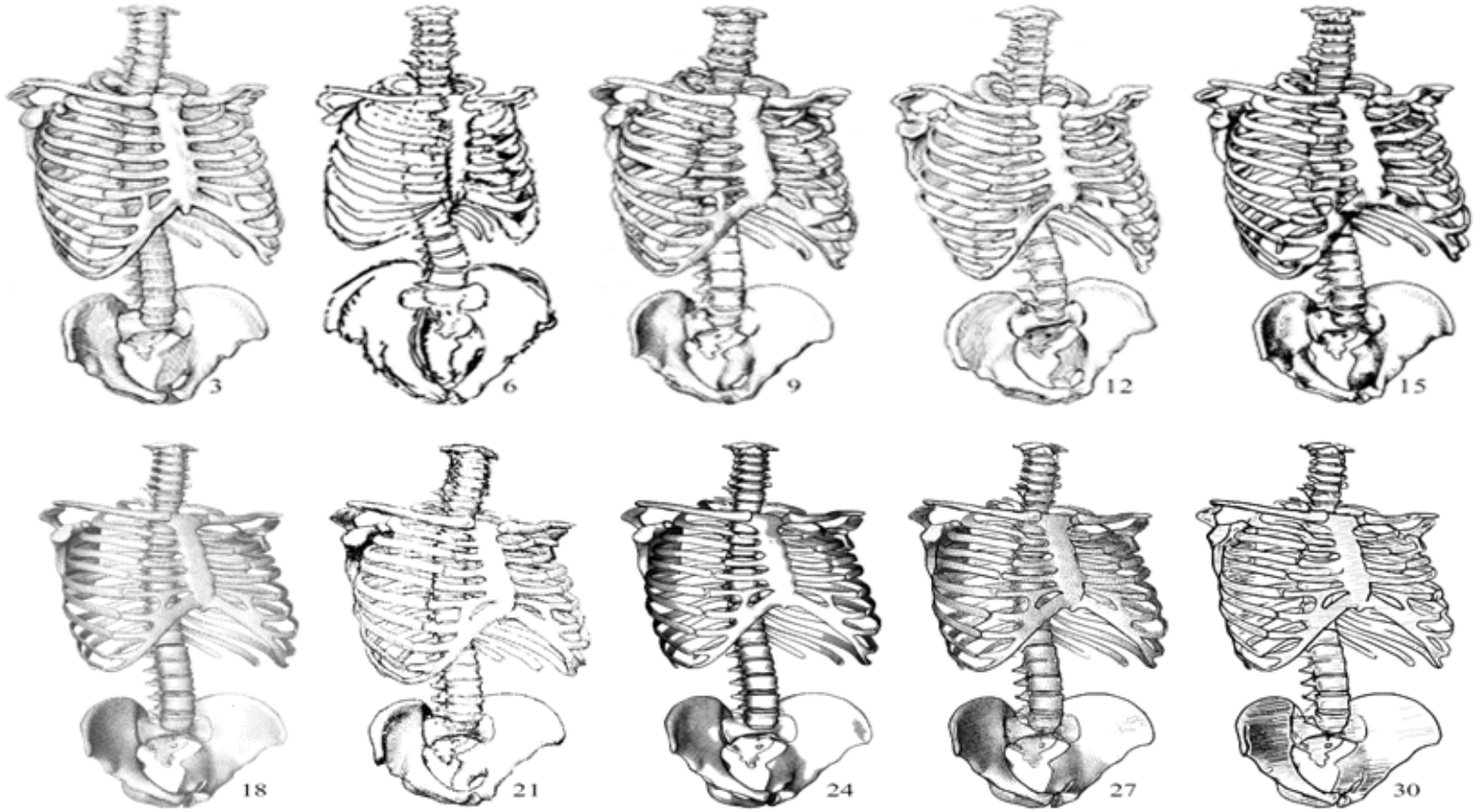
# Abstract Imagery



# Abstract Animation

# Animation Control

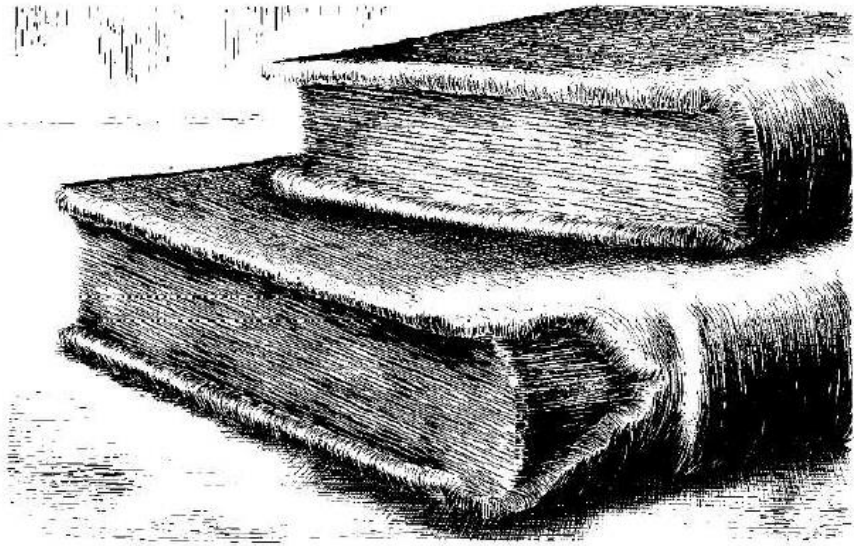
# Simulating Human Styles






© Isenberg et al

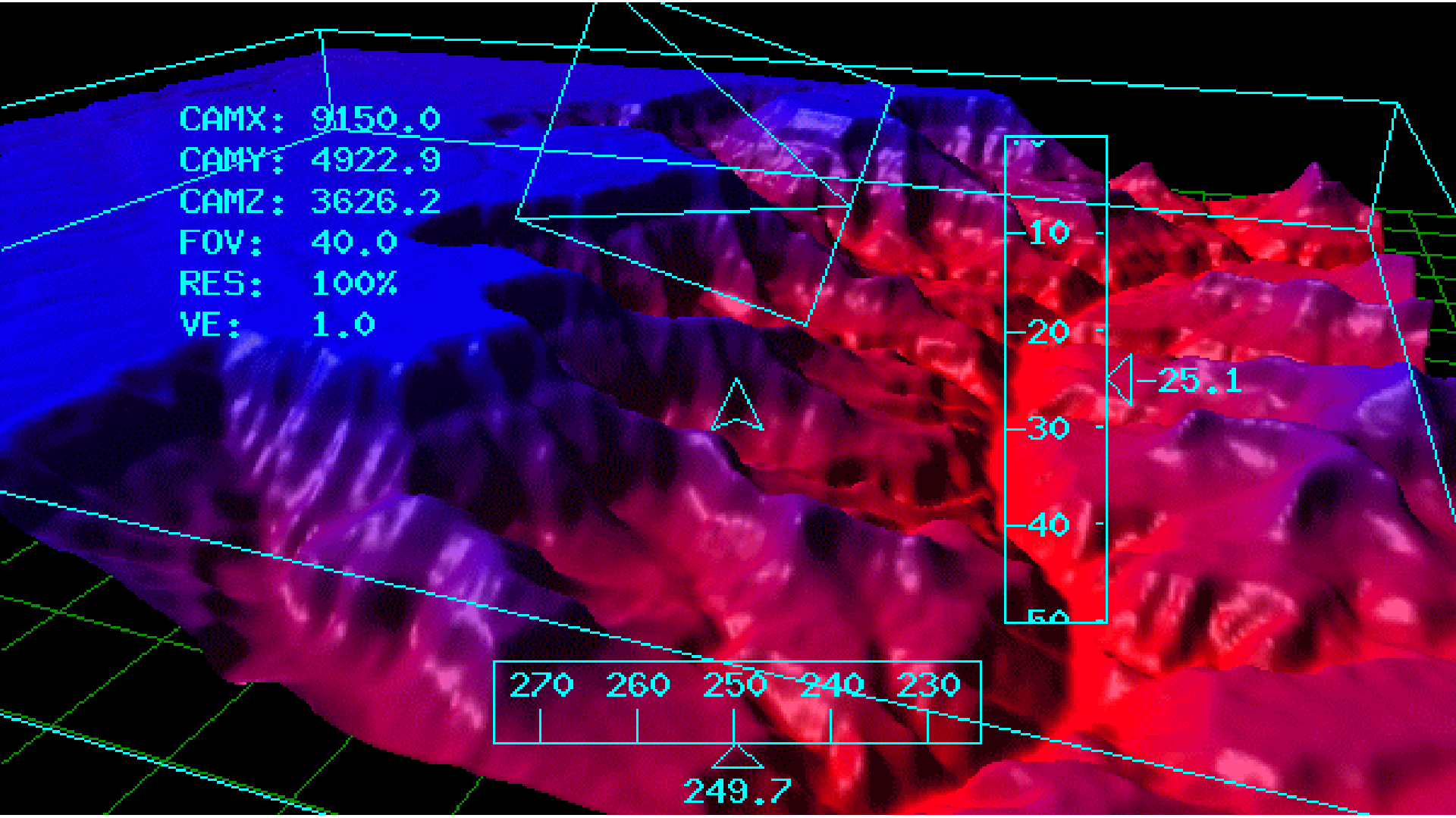
# Simulating Art



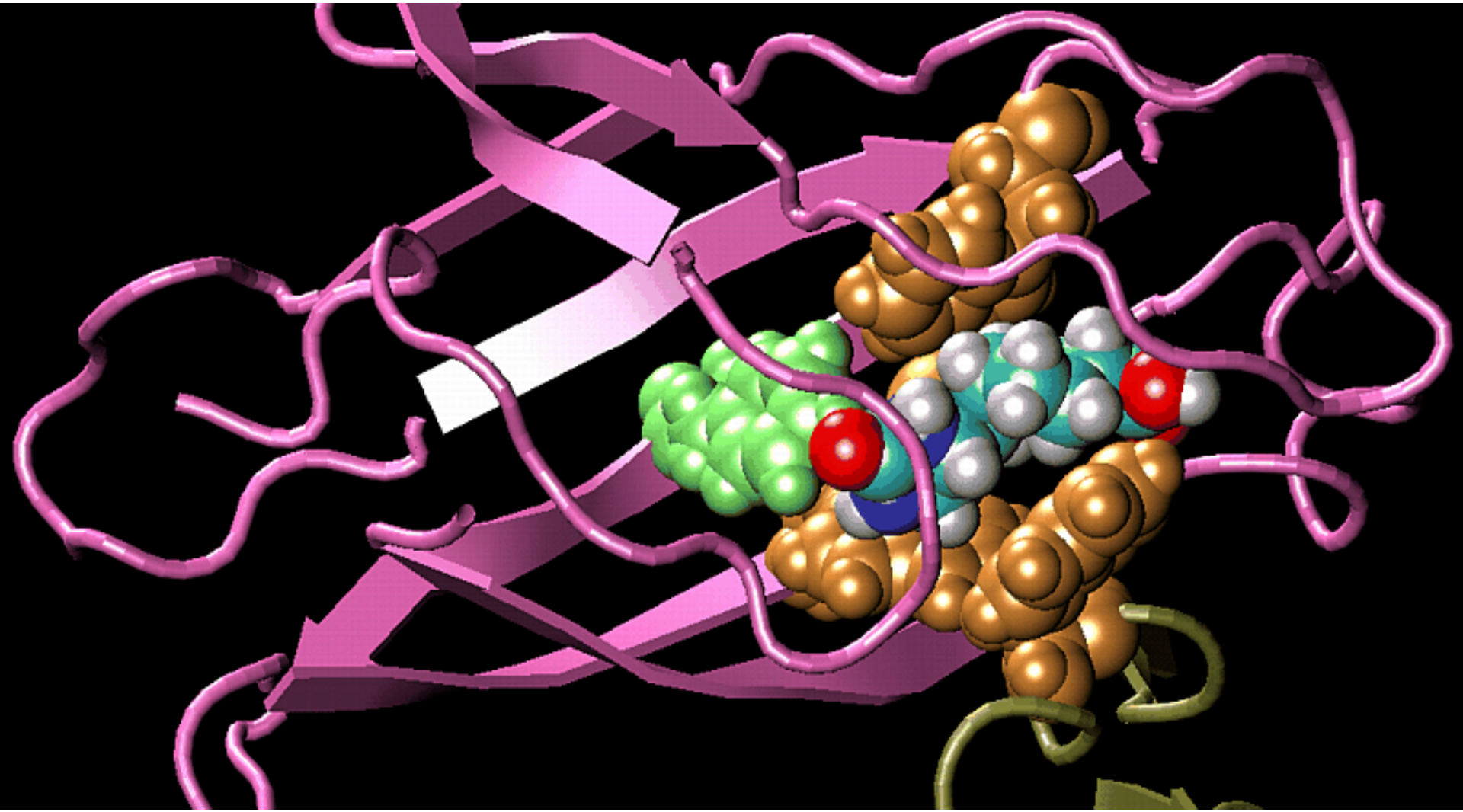
© Salisbury et al [1997]

# Cultural Heritage

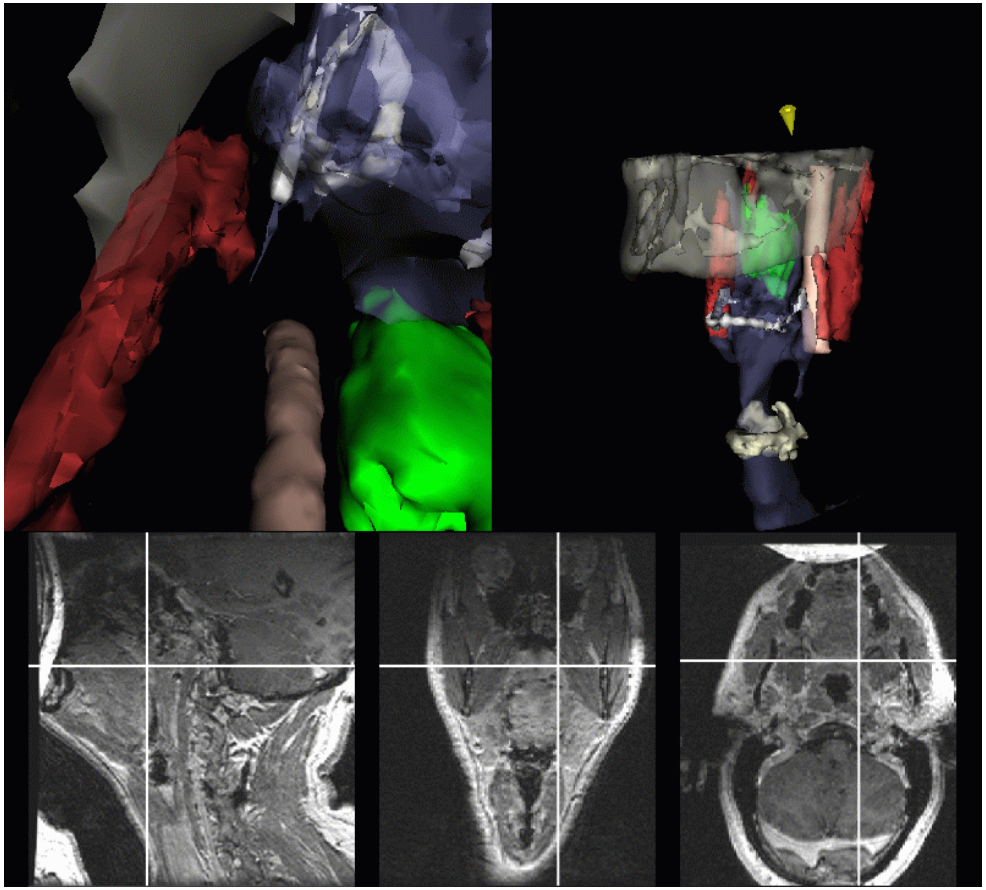
# Scientific Visualization and Analysis



# Scientific Visualisation



# Medicine and Virtual Surgery



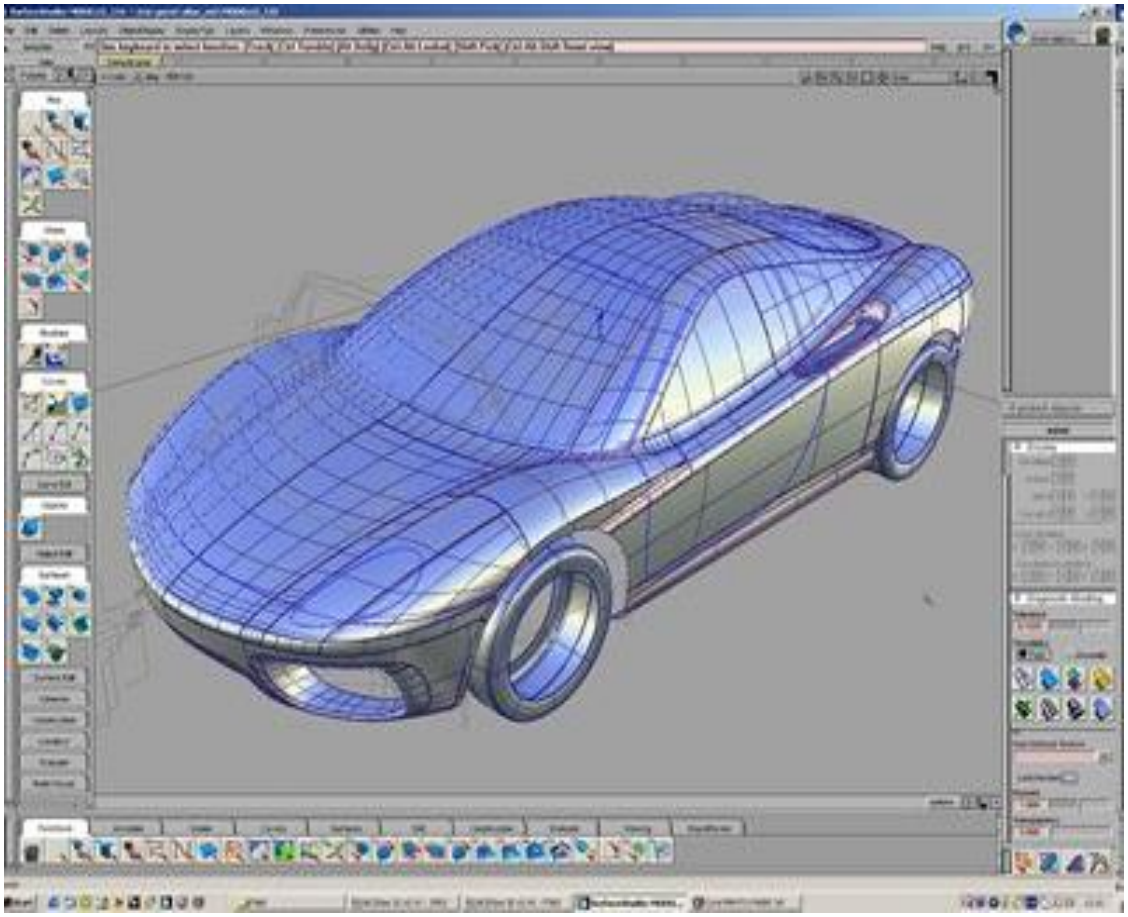
# Lighting and Layout Design

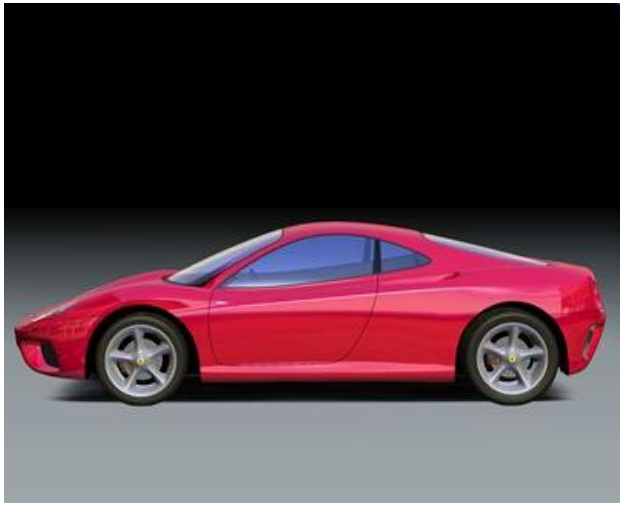
© Chang Kyung Seok

# Architectural Design and Simulation

© NBBJ Architects

# Commercial Design

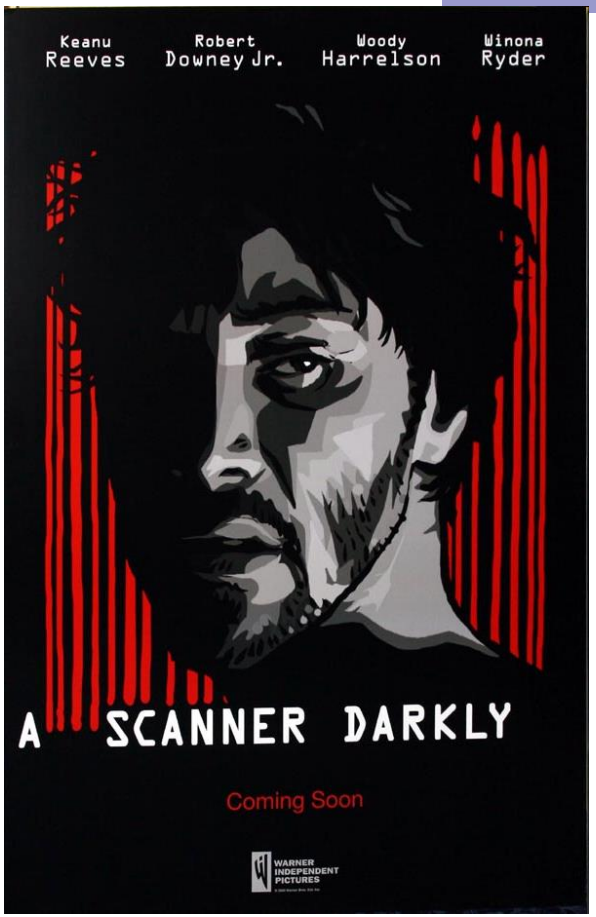




# Image Abstraction

© Anthony Santella and Doug DeCarlo, 2004

# Video Abstraction




© Warner Independent Pictures

# Interaction

# Interactive Games

# Immersive Reality

# Advanced Interfaces

# Augmenting Reality

# Photorealistic (Virtual) Reality

# Difference between AR & VR

**Think scuba diving vs. going to the aquarium**

With virtual reality, you can swim with sharks. And with augmented reality, you can watch a shark pop out of your business card.

# Difference between AR & VR

Augmented reality however, takes our current reality and adds something to it. It does not move us elsewhere. It simply "augments" our current state of presence, often with clear visors. Seen below, Samsung is near ready to introduce its Monitorless AR glasses, which would connect to phones or PCs via WIFI and replace the screen on those devices.

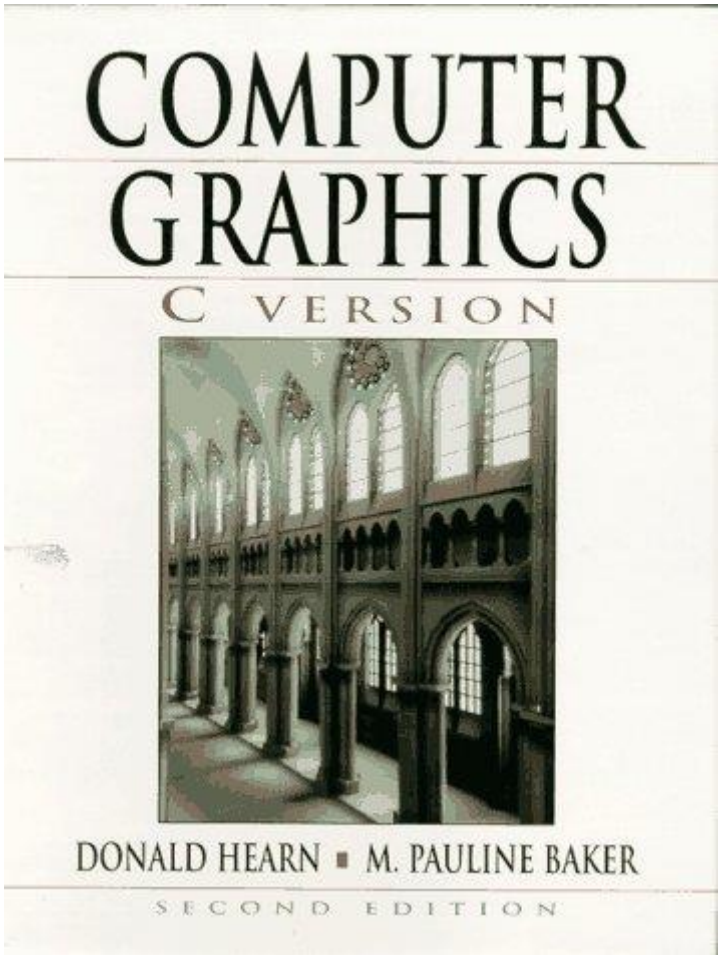
While VR is more immersive, AR provides more freedom for the user, and more possibilities for marketers because it does not need to be a head-mounted display.
https://www.youtube.com/watch?v=SCrkZOx5Q1M

Pepsi Max utilized AR in incredible ways throughout their #LiveForNow campaign. The monster mirror trick is worth noting, but the most amazing display was for their bus stop prank.
https://youtu.be/Go9rf9GmYpM

https://www.youtube.com/watch?v=l3V8zeSljUU

# Textbook

- Computer Graphics
  C Version
  - D. Hearn and M. P. Baker
  - 2nd Edition
  - PRENTICE HALL



# Contact

- Course Web Page
  - *http://202.141.129.18/rosy8/materials3.html*

# The CG Spectrum

- Data: Images and video
- Interactive applications: e.g. Games, Simulators, VR, Walkthroughs
- Tools: Video + image editing, 3D modelling applications
- Scripts, code, bits and bytes, mathematical equations
- Circuits

- Passive Audience (viewers)
- Interactive Audience (users)
- Authors
- Programmers
- Hardware engineers

# Input & Output Devices

- ***Input devices***
  - Keyboard
  - Mouse
  - Trackball & Spaceball
  - Joysticks
  - Data Glove
  - Digitizer/Graphical Tablet
  - Image Sanners
  - Touch panels
  - Light Pens

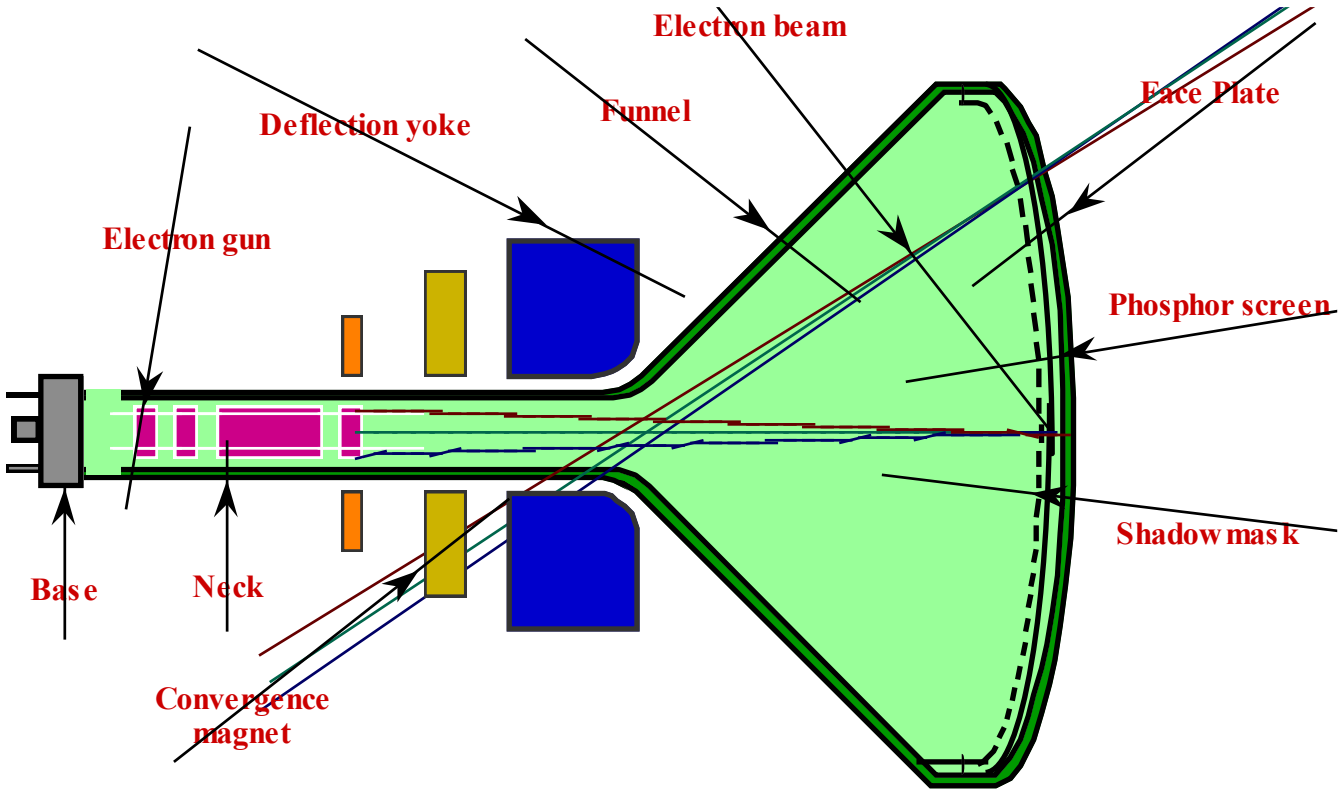
# Input & Output Devices

- **Video Display Devices**
  - Cathode Ray Tubes
  - Vector scan/Random scan Display
  - Raster scan Display
  - color CRT monitors
  - DVST
  - Flat panel Display
  - Plasma panel Display
- **Soft Copy Devices**
- Soft copy
  - CRT
  - Flat panel
    - LCD
    - Electroluminescent
    - Plasma
  - Projector
- **Hard Copy Devices**
  - Printers
    - Dot matrix
    - Ink jet
    - Thermal transfer
    - Laser
  - Plotters

# Video Display Devices

- Operation is based on Cathode Ray Tubes
- One method is to redraw the picture repeatedly by quickly directing the electron beam back over the same points.
- This type of display is called Refresh CRT.

# Cathode Ray Tube



# Electron Gun

- Heating Filament
- Cathode
- Control grid
- Focusing Anode
- Accelerating Anode
- Electron beam path

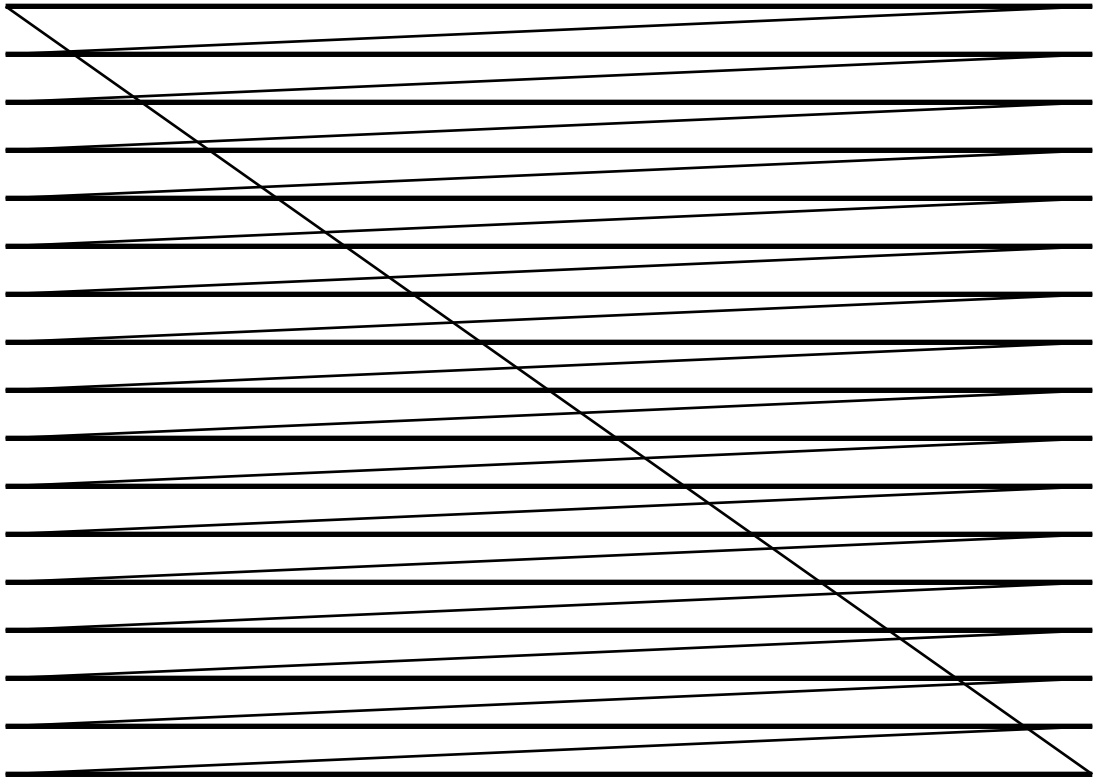
# Raster Scan Display

- It is based on Television Technology
- The electron Beam swept across the screen one row at time from top to bottom.
- The picture definition is stored in a memory area called “refresh buffer” or “frame buffer”
- Each screen point is called as “pixel”

# Cathode Ray Tubes

| Advantages | Limitations |
| --- | --- |
| Very LOW cost | Large volume/weight |
| Good visual quality | Size < 45 inches |
| Wide viewing angle | Repeatability |
| Simple grayscale | High voltage |
| Simple addressing | |
| Large manufacturing base | |

# CRT: Raster Scan



# Concepts

- Bitmap
- Pixmap: A system with 24 bits/pixel & a screen resolution of 1024x1024 requires 3 MB of storage for refresh buffer.
- Aspect Ratio: vertical points/ horizontal points necessary to produce equal length lines in both direction on the screen
- Refreshing on raster scan displays 60 to 80 frames per second
  - Horizontal, vertical retrace & interlacing
- Major difference between phosphors
  - Persistence
  - Resolution
  - color

# Random Scan or vector displays

- Draws a picture one line at a time
- Refresh rate depends on number of lines to be displayed
- Picture definition is stored as a set of line-drawing commands in an area of memory i.e., the refresh display file
- After all line drawing commands are processed the system cycles back to the 1st line command in the display file
- Draws all the component lines of a picture 30-60 times / second
- For small set of lines refresh cycle is delayed to avoid refresh rates > 60 frames/ sec

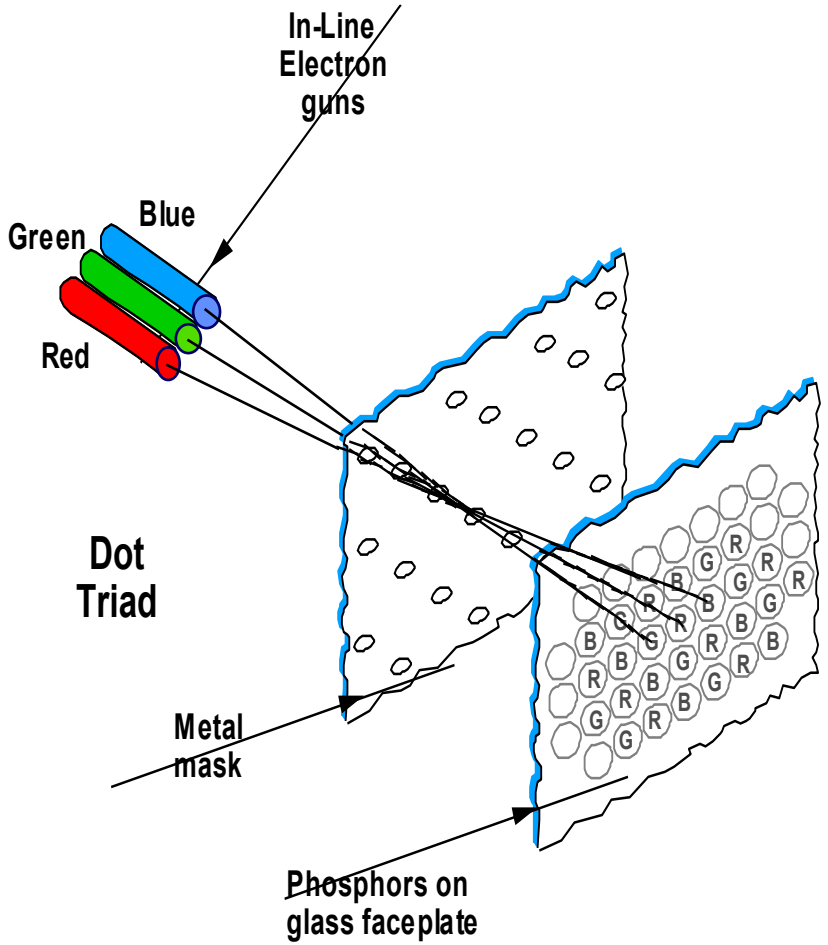
# Color CRT monitors

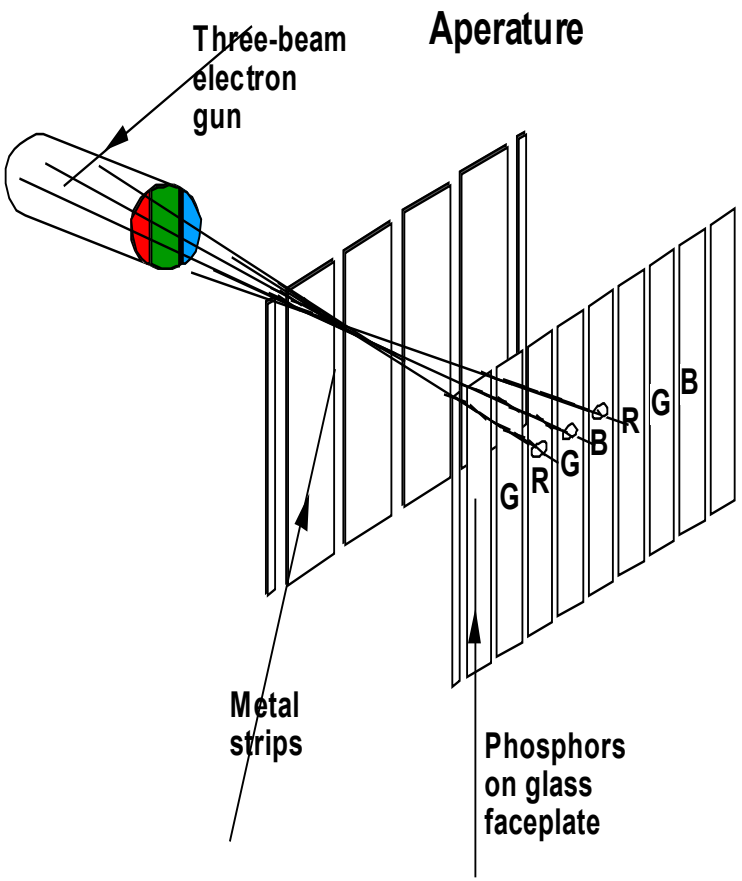
- Displays color pictures by using combination of phosphors that emit different-colored light
- Basic Techniques for producing color displays:
  - Beam-penetration method: 2 layers of phosphor, red & green, are coated onto the inside of the CRT screen; displayed color depends on how far the electron beam penetrates into the phosphor layers
  - Shadow-mask methods: delta-delta method & the in-line arrangement

# Color CRT Monitor: Shadow-mask

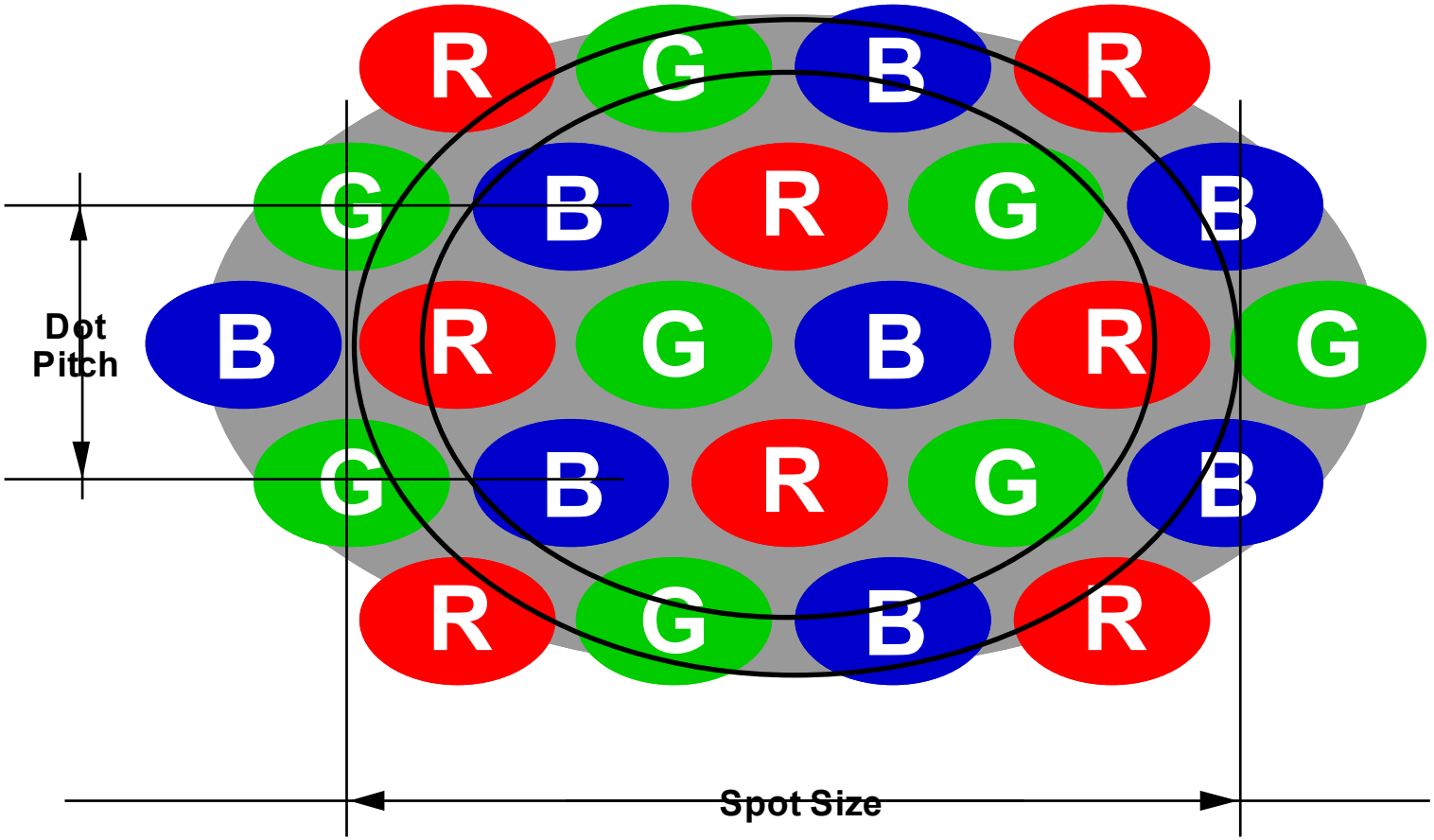
- Colored pictures can be displayed by using a combination of phosphorous that emit different colored light
- Three phosphor color dots at each pixel position.
- This type of CRT has three guns, one for each dot, and a shadow mask grid just behind the screen.
- These dots are arranged either triangular pattern or in an inline pattern along a scan-line.
- Various color intensities can be obtained by varying the intensity of three beams.

# Color Shadow Mask CRT



# Color CRT Phosphor Pattern Versus Spot Size



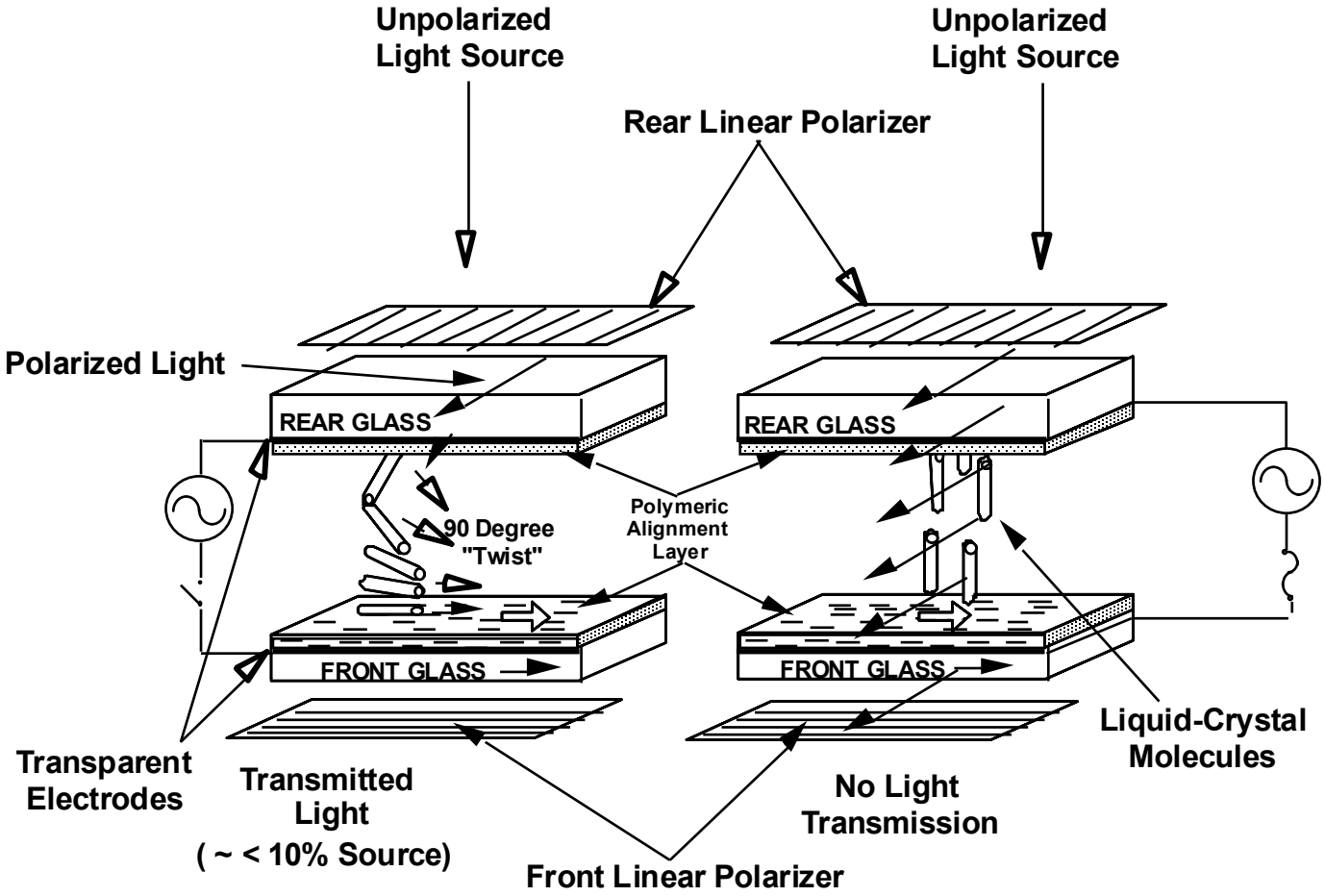
# Flat Panel Display

- A class of video devices that have reduced volume,weight &power requirement than refresh CRT.
- Currently they are used in calculators , pocket video games, laptop computers etc.
- There are two categories
  - Emissive displays: convert electrical energy to light. eg. Plasma panel, LEDs etc
  - Non-emissive displays: use optical effects to convert sunlight or light from other sources into graphics patterns. eg. LCD

# Output Devices

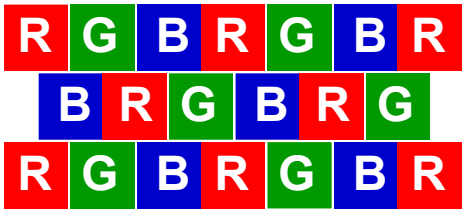
- Hard copy
  - Printers
    - Dot matrix
    - Ink jet
    - Thermal transfer
    - Laser
  - Plotters
- Soft copy
  - CRT
  - Flat panel
    - LCD
    - Electroluminescent
    - Plasma
  - Projector

# Liquid Crystal Display (LCD)



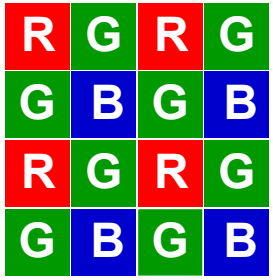
# Flat Panel Pixel Arrangements

**Triad**




**Stripe**

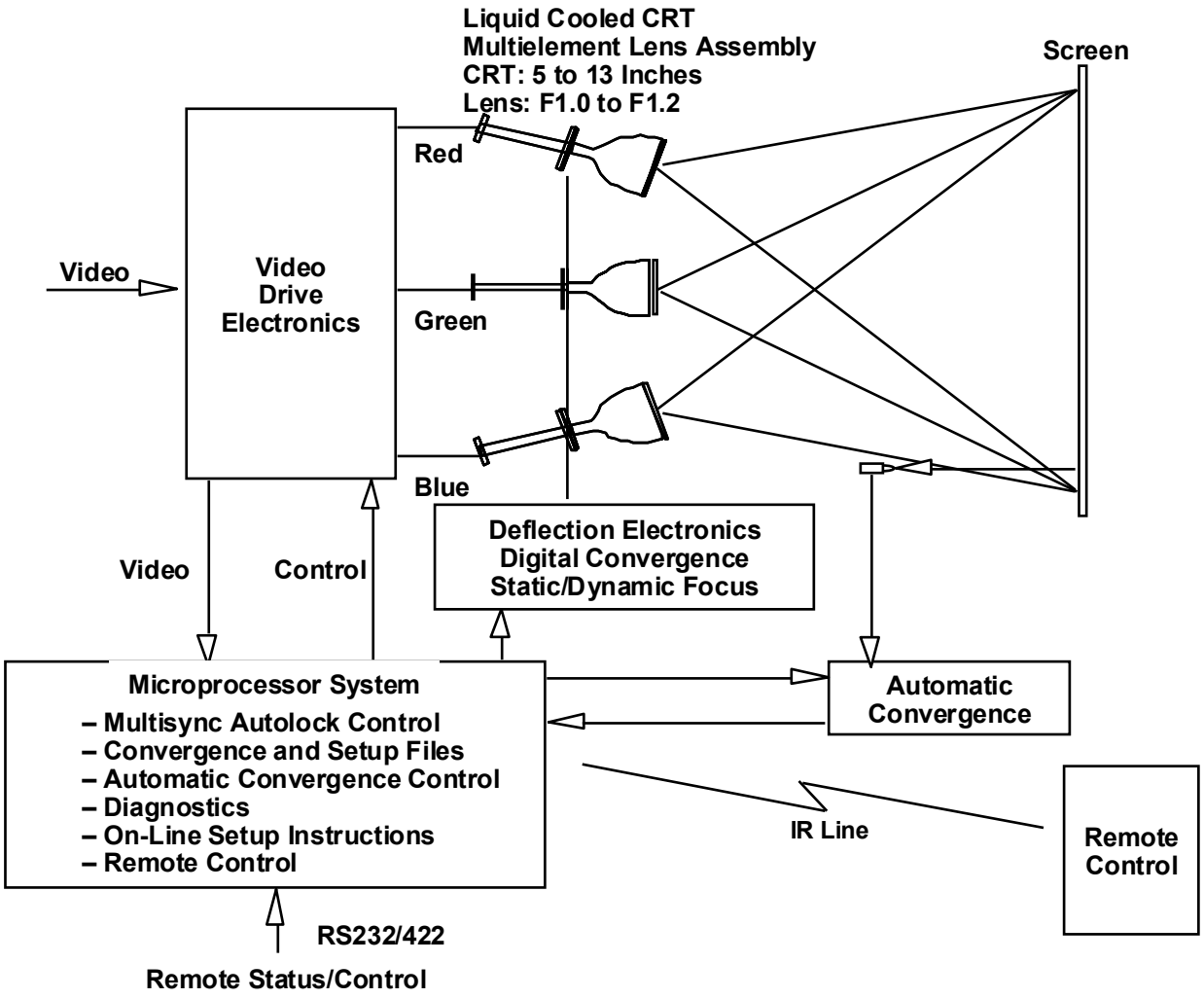
**Quad**



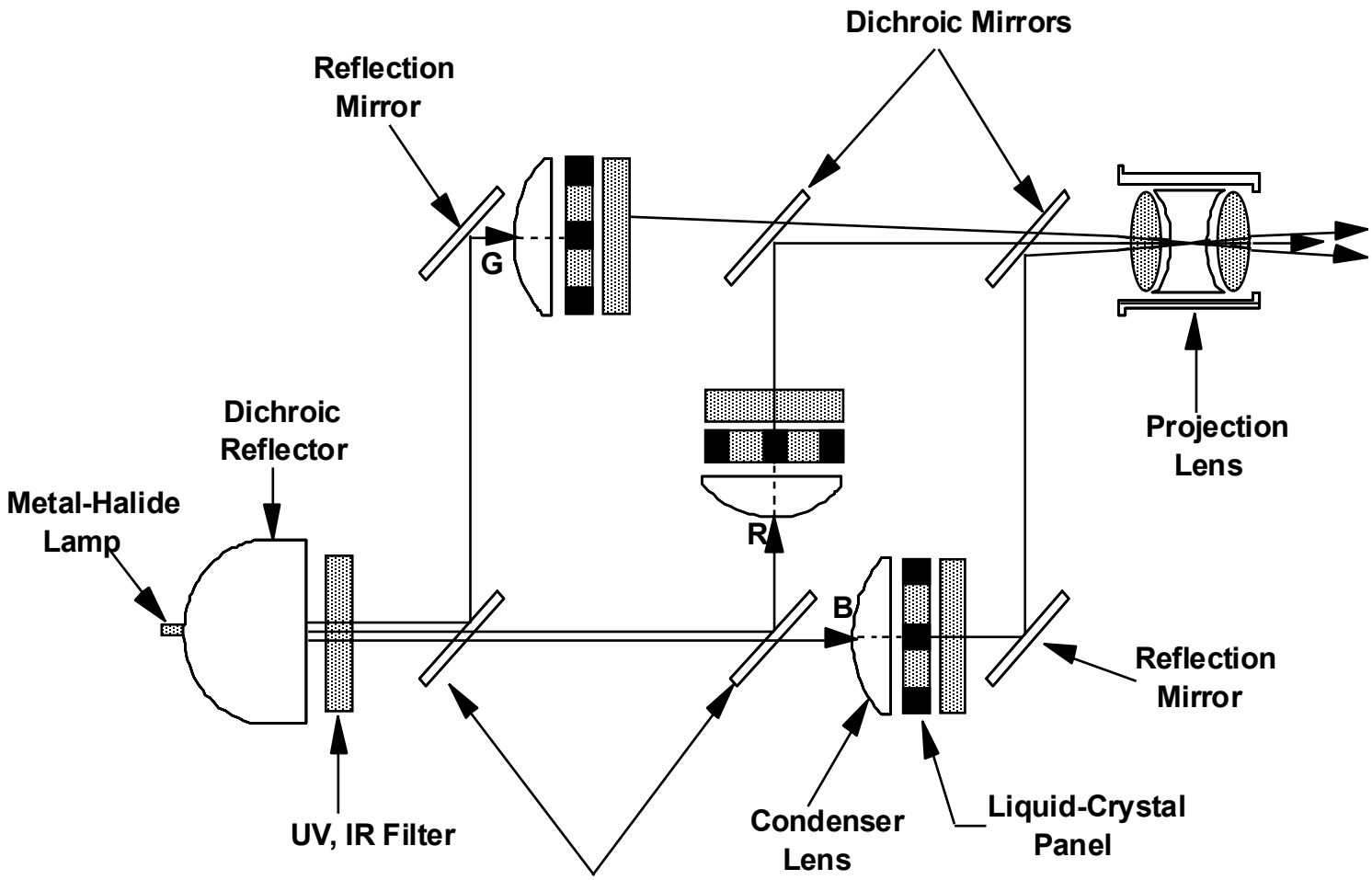
# *Liquid Crystal Displays*

| Advantages | Limitations |
| --- | --- |
| Thin/low weight | Image size < 13 inches (passive) |
| Rugged and reliable | < 16 inches (active matrix) |
| Low voltage drivers | Response time (passive) |
| | Premium cost (active matrix) |
| | Limited viewing angle |
| | Limited temperature range |

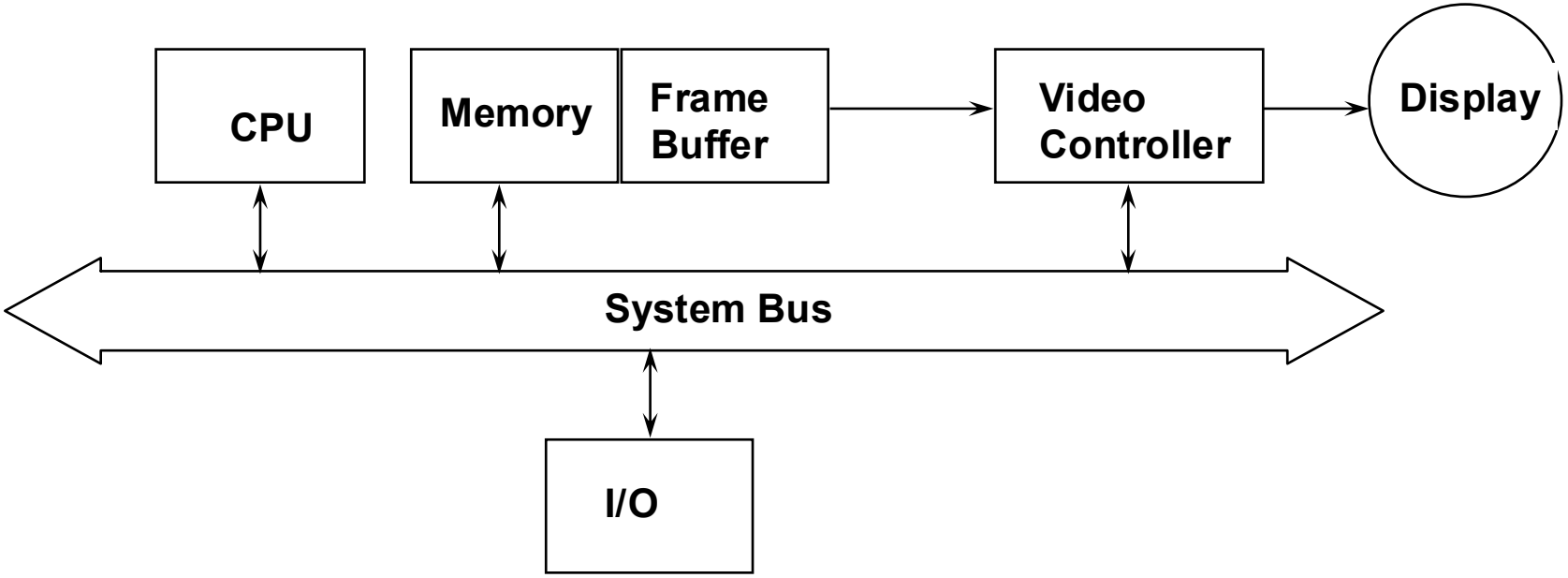
# CRT Projection System



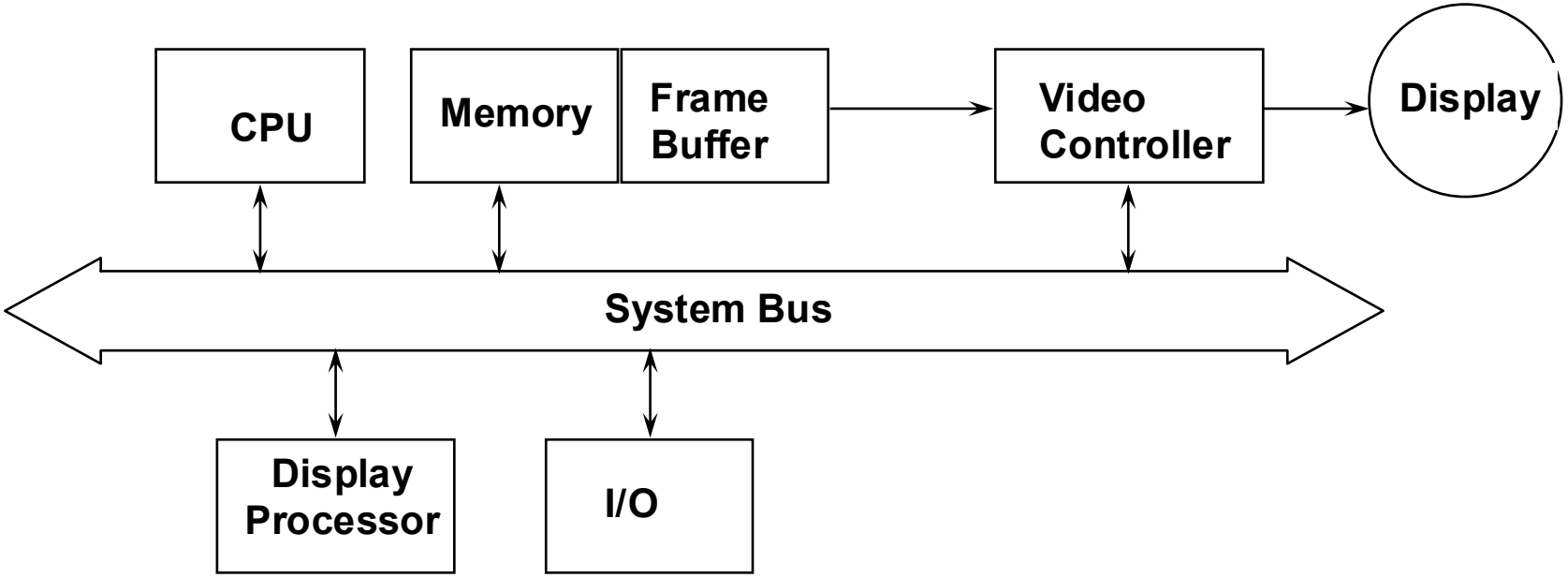
# LCD Projector



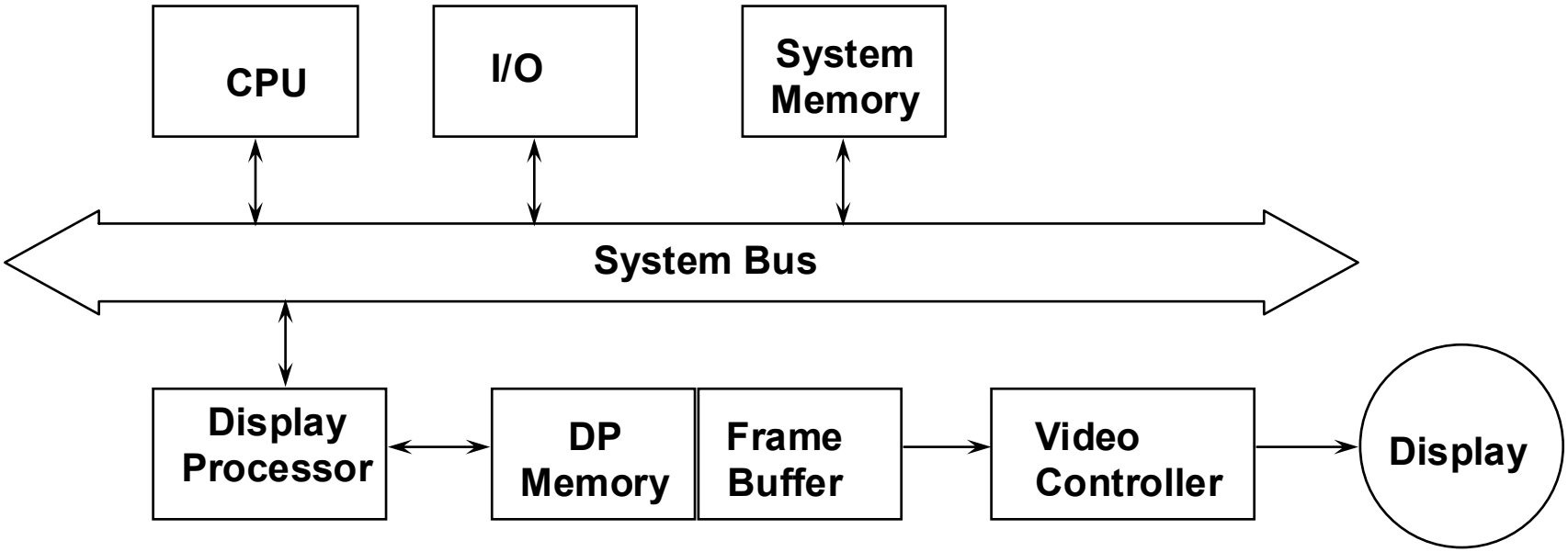
# Common Raster Display System Architecture



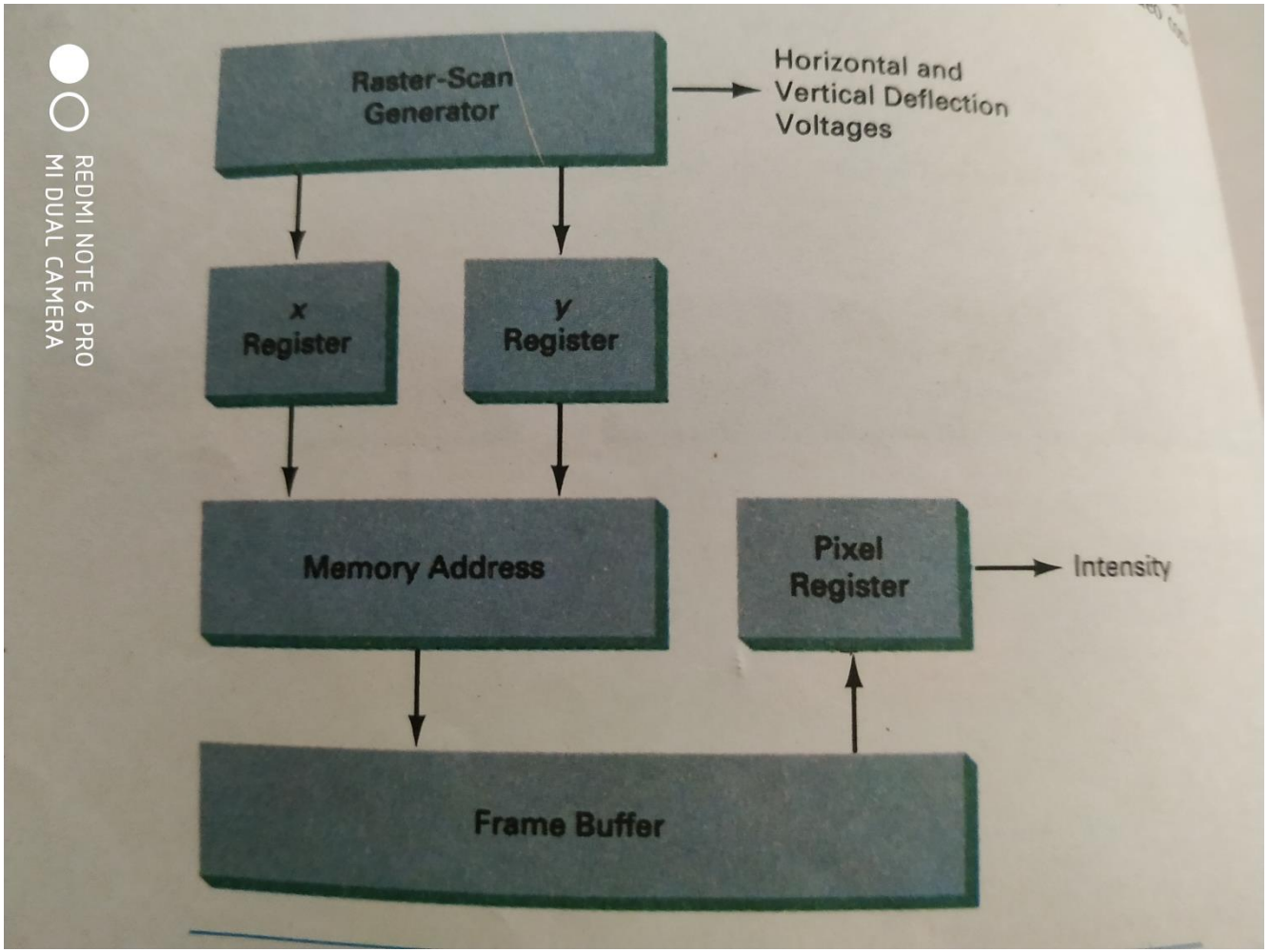
# Common Single Address Space Raster Display System Architecture



# Raster Display System Architecture With Peripheral Display Processor



# *Video controller refresh operations*

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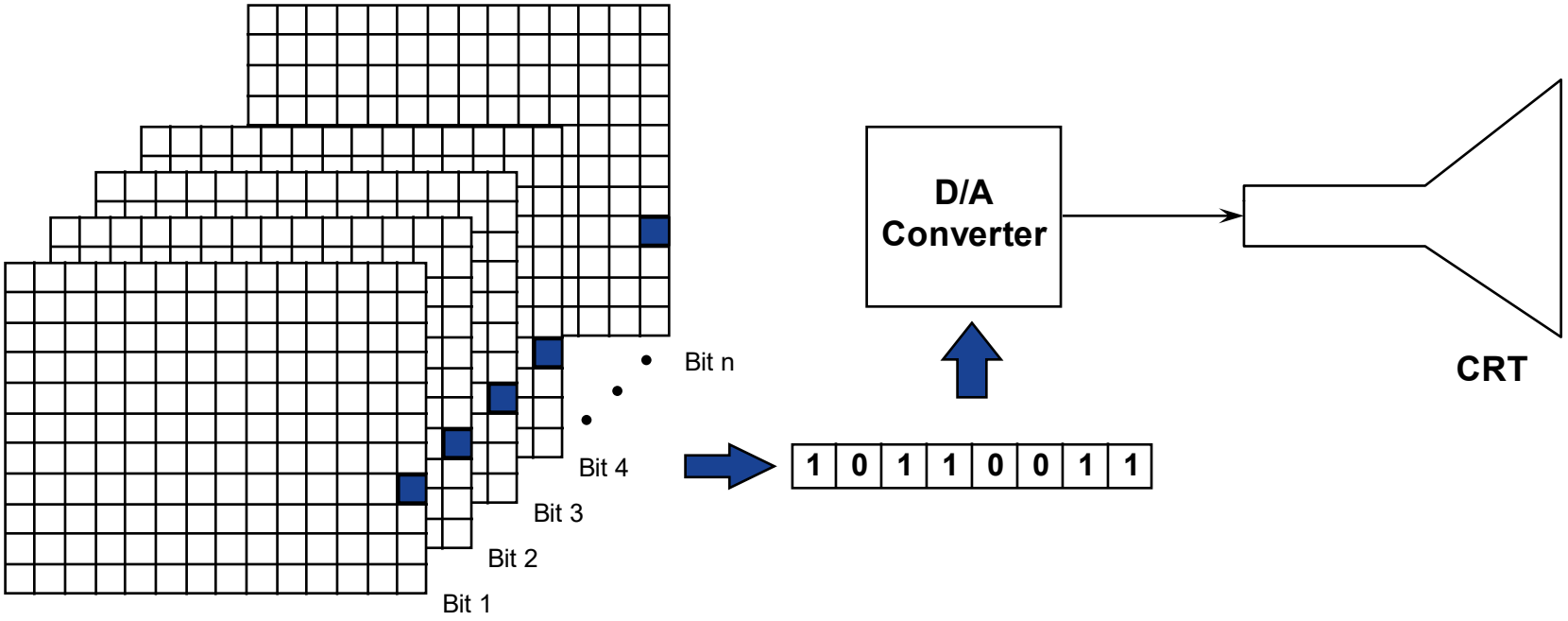
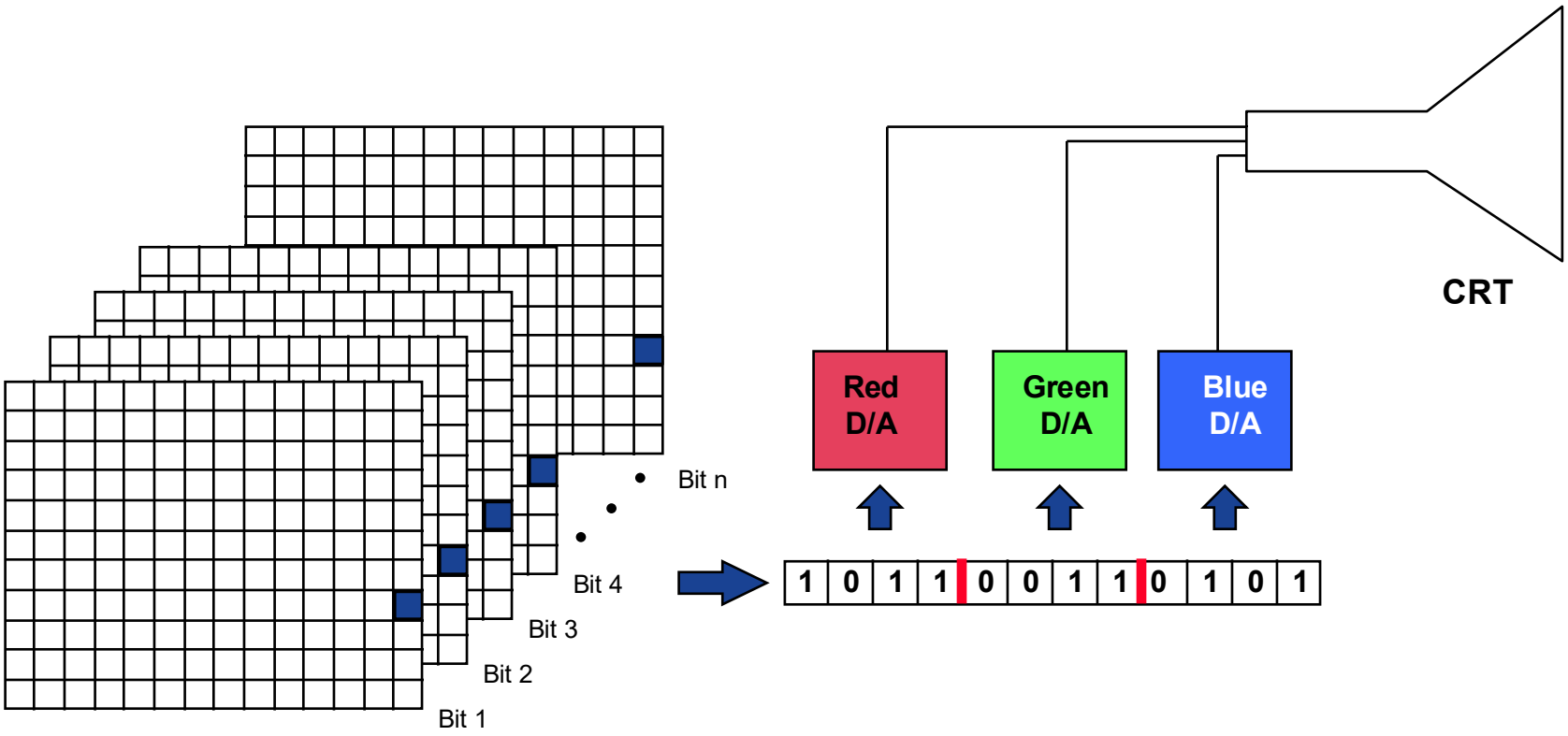
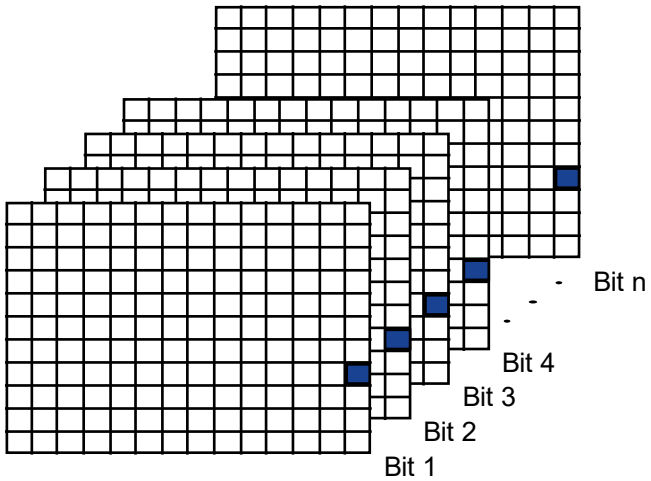
# Monochrome Frame Buffer

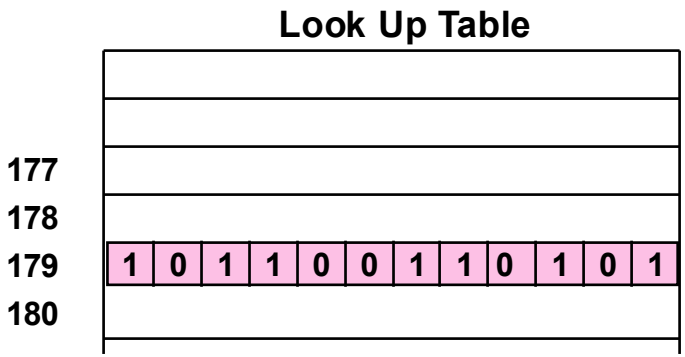
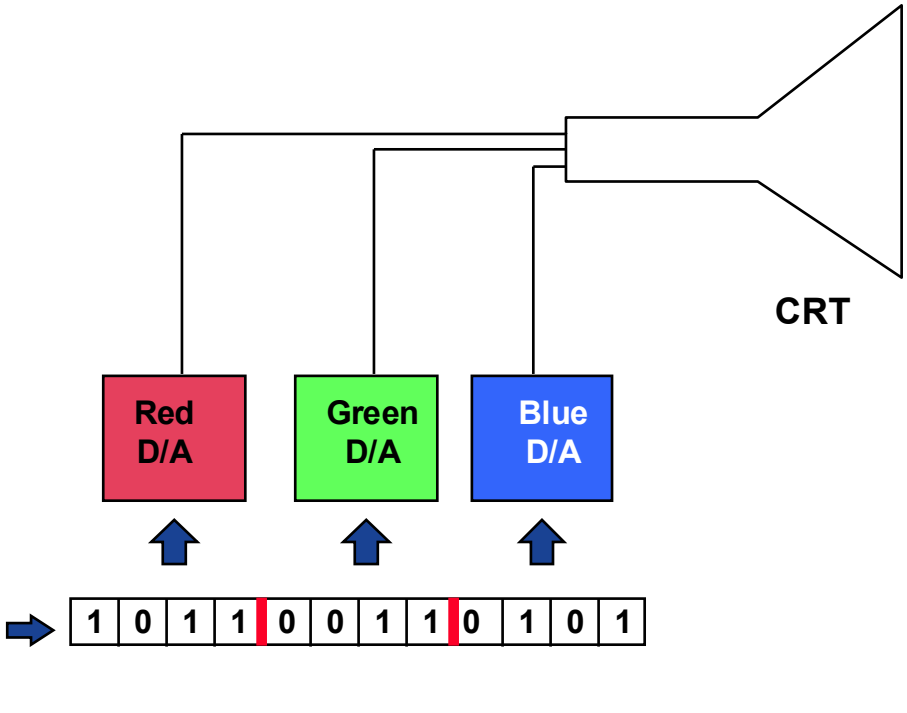


# Color Frame Buffer



# Video Look-up Table




| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
|---|---|---|---|---|---|---|---|

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